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Reusing Existing Infrastructure for CO₂ Transport: Risks and Opportunities

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Abstract

There is no doubt that greenhouse gas emissions, particularly CO₂, needs to be reduced to mitigate the effects of climate change. While carbon management can be achieved through a number of technological and engineering approaches ranging from energy efficiency (i.e., highly energy integrated system and process intensification) to renewable energy (wind, solar, hydrogen), CO₂ capture & storage (CCS) has been identified as having a key role in the energy transition.

Captured anthropogenic CO₂ can be permanently stored in saline aquifers and depleted reservoirs. Saline aquifers (normally unsuitable for industrial or human exploitation) offer the largest storage capacity; however, there is, usually, lack of geological characterization leading to high risks due to large uncertainty. On the other hand, depleted gas fields, close to economical life cessation, are deemed an excellent alternative as safe and long-term storage is already proven and immense geological characterisation has been gathered during production life. Moreover, there is great potential to repurpose the existing offshore infrastructure (pipelines, platforms, and wells) as to minimize capital expenditure and delaying decommissioning costs. Repurposing existing production systems can also be an efficient way to achieve rapid deployment of CCS at large scale.

In this paper, we present the key engineering challenges, risks, and opportunities in the re-use of existing oil and gas offshore infrastructure for CO₂ transport and injection. We highlight the complex operational constraints and interactions between different components of the transportation network. The design and operation of the transportation network is governed by the following drivers:

- Safe design
- Robust and flexible operation
- Minimize cost (or delay expenditure as long as possible)
- Minimize emissions of greenhouse gases associated to the operation of the transport network (i.e., energy efficiency)
- Start operation with minimum modifications

Keywords: CCS, CO₂ transport, Flow Assurance, pipeline repurposing, infrastructure, carbon management

Introduction

As more countries (and industries) pledge net-zero targets to meet the climate commitments made at the United Nations COPs of Paris and Glasgow, it is recognized that Carbon (CO₂) Capture and Storage (CCS) will play a key role, at least in the short-term, in achieving such ambitions, e.g. [Lau et al., 2021](#). It is suggested that CCS will be a bridge to transition, in the coming decades, to a, hopefully, largely decarbonized economy with the associated transfer and creation of, new, skills and jobs.

CCS is a process that separates CO₂ from a flue stream (e.g., from combustion of fuel in an industrial process such as power plants, cement kilns, steel manufacturing, etc.) or from acid gas (e.g., from natural gas with high CO₂ content). Then, the captured CO₂ is processed (purified and dried) for transport and injection into suitable geological formations (i.e., depleted reservoir, saline aquifers, deep unminable coals seams) for permanent storage. Also, CO₂ has been widely utilized for various decades to extend hydrocarbon production (e.g., Enhanced Oil Recovery (EOR) ([Orr and Taber, 1984](#); [Holm, 1982](#))).

CCS consists, typically, of the following main operations: separation or capture (e.g., pre-combustion ([Jansen et al., 2015](#)), oxyfuel ([Stanger et al., 2015](#)), post-combustion ([Liang et al., 2015](#); [Chao et al., 2021](#)), acid gas removal from hydrocarbons ([Qayyum et al., 2020](#)), capture from industrial processes ([Bains et al., 2017](#)) or direct capture from air ([Sanz-Pérez et al., 2016](#))), conditioning (purification and dehydration to meet a specification ([Abbas et al., 2013](#); [Kemper et al., 2014](#))), compression ([Kumar et al., 2010](#); [Modekurti et al., 2017](#); [Jones and McKaskle, 2014](#)), transport (onshore & offshore pipelines ([Roussanaly et al., 2013](#); [Roussanaly et al., 2014](#); [Peletti et al., 2018](#)) or transfer via ships ([Decarre et al., 2010](#)), well injection ([Bai et al., 2015](#); [Eiken et al., 2011](#); [Duguid et al., 2018](#)) and, storage ([Michael et al., 2010](#); [Ringrose, 2020](#); [Ajayi et al., 2019](#)). The scope of this paper is related to the transport and injection infrastructure.

At the time of writing this manuscript (late 2021/early 2022), there are no large-scale commercial CCS operations in the world[1]. Though, there are few CO₂ disposal facilities from natural gas production, using CCS, currently operating (e.g., Sleipner in Norway, Gorgon in Australia, CNOOC in China). Similarly, operational large-scale industrial CCS projects are limited (Decatur CCS from ethanol production in the USA and Orca Direct Air Capture in Iceland). A recent summary, up to 2021, of CCS projects can be found in [Loria et al \(2021\)](#). All these projects consist of a single source/emitter exporting CO₂ to a dedicated geological storage using new transport and injection facilities.

To maximize economies of scale, new CCS systems are being developed as industrial ‘clusters’ or ‘hubs’ ([Brownsort et al., 2016](#); [Kahlke et al., 2010](#); [Brunsvold et al., 2011](#)) where emitters of diverse industries (cement, fertilizer, steel, refinery, power, H₂ production, etc.) will be exporting their captured CO₂ to a common shared transport infrastructure (see [Figure 1](#)). Northern Lights (in construction as in 2021) is an example of CCS transport and storage infrastructure which will receive CO₂ emissions from various industrial emitters via ship to a temporal terminal and then further offshore transport via new pipeline for injection into saline aquifer. Other industrial clusters in advanced stages of design (at end of 2021) are HyNet North West (UK) and Porthos (the Netherlands). Both CCS networks transport CO₂ from various sources via onshore and offshore pipelines for storage in depleted gas reservoirs. In the case of HyNet, the existing pipeline infrastructure will be repurposed for CO₂ service ([Becker et al., 2021](#)).

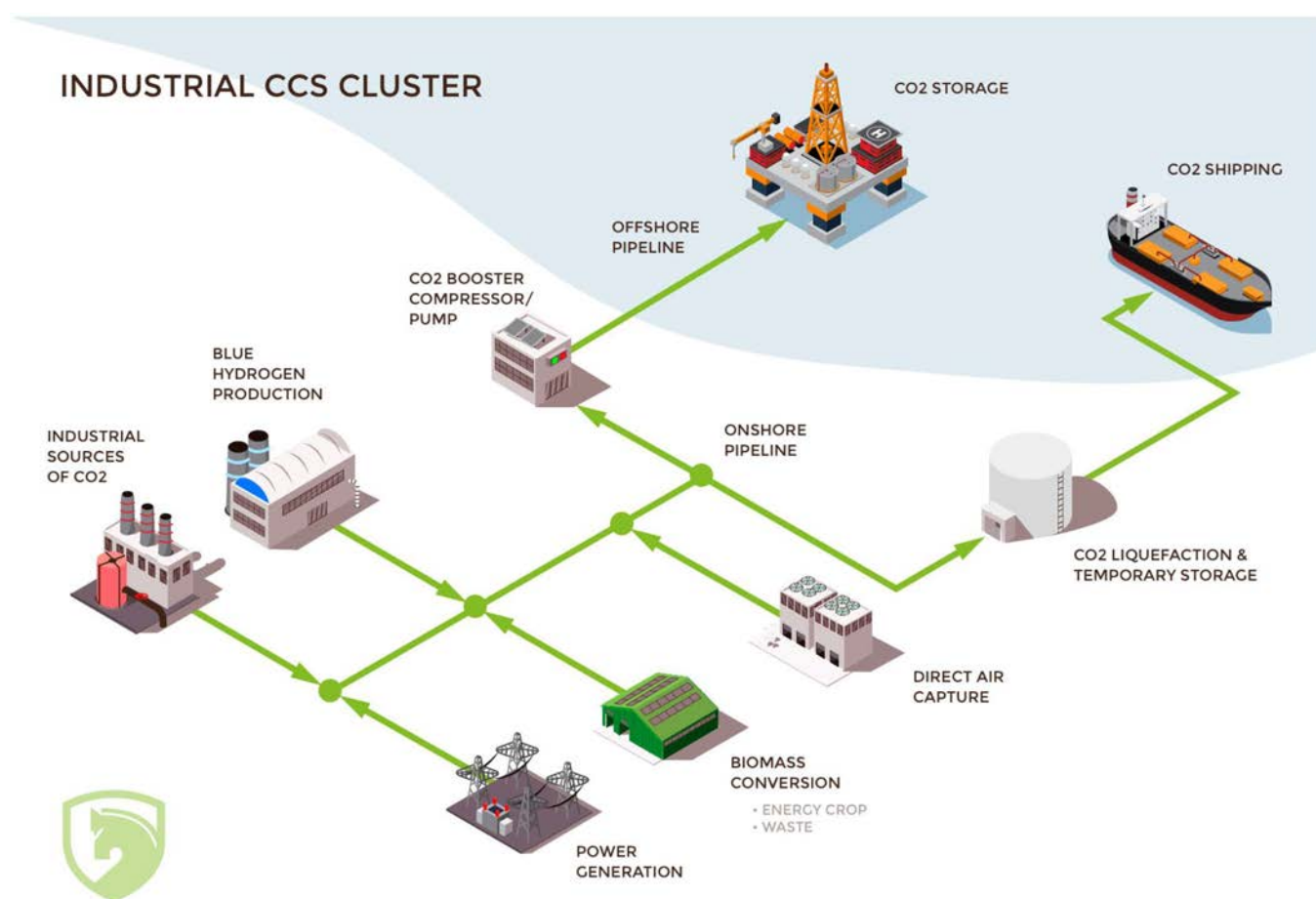


Figure 1—Concept of an Industrial CCS Cluster (with permission of Pace CCS Ltd).

Depleted gas fields, close to economical life cessation, are deemed excellent geological stores because long-term safe storage is already proven, and immense geological knowledge has been acquired during production life. Moreover, there is great potential to repurpose the existing infrastructure (pipelines, platforms, and wells to some extent) as to minimize capital expenditure and delaying decommissioning costs. Repurposing existing production systems could also be an efficient way to achieve rapid deployment of CCS at large scale.

CO₂ transport in pipelines is relatively straightforward and, as discussed earlier, it has been done for many years for onshore EOR, mainly in North America. Indeed, there are lessons learned (Parker et al., 2009; Noothout et al., 2014; Sweatman et al., 2009; Gale and Davison et al., 2004) that could and should be applied in the design of industrial CCS networks. However, CO₂-EOR pipelines are new (onshore) purpose-built which means they can be designed, constructed, and operated to the specific requirements of the fluid. On the other hand, existing natural gas and oil pipelines could be converted to transport CO₂-rich fluid. The aim of this paper is to review and discuss the main technical challenges that repurposing pipelines for CO₂ transport pose.

Storing CO₂ in depleted gas reservoirs

Despite their reduced storage capacity (in comparison to saline aquifers), depleted (or depleting) gas reservoirs offer excellent opportunities to permanently store CO₂. Firstly, reservoir operators have accumulated, potentially decades of, considerable amount of geological information (pressure, permeability, rock integrity, seismic, porosity, etc.) during the life of field: static and dynamic characterization is known (new appraisal or exploration campaigns might not be required!). Prior to CO₂ injection, there will be a

clear picture of the storage capacity, pressure build-up response and, in general, of any limitations of the reservoir. In addition, long-term containment is certain as they have seals that preserved gas for millions of years. Importantly, as many fields worldwide are closed to their end of economic life, the depleted reservoirs can be readily available in short-term allowing, potentially, rapid large-scale deployment of CCS hubs. In [Loizzo et al. \(2009\)](#), a detailed discussion on the advantages of depleted gas fields for CCS (from the point of view of storage capacity, injectivity and containment) can be found.

To date (end of 2021), there are no commercial CCS facilities storing to a depleted gas reservoir. However, few small-scale demonstration projects have been developed with successful results, lessons learned, and risks identified and assessed ([Hannis et al., 2017](#)). Here, the following small subset of demonstration projects can be mentioned: Otway, onshore Australia, which in its first phase stored circa 65,000 t of CO₂ reporting no safety issues and monitoring results that confirm good understanding of the storage process ([Jenkins et al., 2012](#)). The initial "depleted" bottom-hole pressure is very high 178 bar with a bottom-hole temperature of 63°C. K12-B, offshore Netherlands, had 15 years of CO₂ injection before its conclusion in 2017 ([Vandeweyer et al., 2021](#)). The general behavior of the field observed could be explained and, in some cases, even predicted. It is understood that depleted reservoir pressure (prior to injection) was 40 bar with a considerable high temperature of 127°C ([Vandeweyer et al., 2011](#)). Lacq-Rousseau (onshore France) stored 51,000 t before its closure in 2015 ([Hannis et al., 2017](#)). The project made use of existing pipeline that originally transported natural gas. The initial depleted pressure of the reservoir was 39 bara with a high temperature of 150°C ([Monne and Prinnet, 2013](#)). No operational issues were observed from a well performance or injectivity. In addition, the project demonstrates a full-integrated chain operation comprising of oxyfuel capture, compression and drying, transport and injection. It follows that the using depleted stores to sequester CO₂ have been demonstrated as feasible.

By the mid 2020's, few large-scale industrial CCS hubs/cluster are planned to start operation (mainly in Europe, where CCS has rapidly evolved) injecting into depleted reservoirs: the previously mentioned HyNet (UK), Porthos (the Netherlands), and East Coast Clusters (UK) with many others to come (Acorn (UK), V Net Zero (UK), Morecambe Bay (UK), Greensand (Denmark), Aramis (the Netherlands), to name just a few that are at different engineering stages).

In Southeast Asia, the Kasawari CCS plans to capture CO₂ from sour natural gas production for injection into a depleted reservoir offshore Sarawak. It is understood that the long-term plan is to develop an industrial CCS cluster. Similarly, the Lang Lebah development will include CCS facilities, for natural gas processing, as part of their CO₂ disposal process into depleted reservoirs. In Indonesia, the Kaliberbau Dalam onshore gas discovery will include CO₂ capture for injection in depleted fields (Dayung and Gelam). In South Korea, a demonstration project for industrial CCS, reusing depleted reservoirs in the Donghae gas field, is currently being considered. The Gulf of Mexico Partnership for Offshore Carbon Storage (GoMCarb) is currently studying options for industrial hubs with potential to reuse existing infrastructure (onshore and offshore) ([Sachde et al., 2019](#)). This is a small sample of developments which, at the time of writing, are at early stages of engineering design.

While there is clear potential and opportunities to exploit depleted reservoirs to store CO₂, there are also risks. Demonstration projects have identified risks and addressed some of them (see for example [Hannis et al., 2017](#)). Undoubtedly, the key challenge is to preserve the integrity of the reservoir. In [Hughes \(2009\)](#), key issues associated to the injection of CO₂ into depleted reservoirs are reviewed. Amongst these risks, the effects of excessive Joule-Thomson (JT) cooling in the reservoir stands out as paramount.

As fluid is being injected into the reservoir, the CO₂ will experience a pressure drop across the formation (near well-bore). This pressure drop will lead to JT cooling as the fluid expands. The main concerns ([Oldenberg, 2007](#)) are, firstly, that the fluid will cool down excessively such that there is a risk of hydrate formation and freezing of the residual brine trapped in the pores leading to loss of injectivity. Secondly, the potential impact into the integrity of the rock as it experiences extreme thermal stresses due to the cooling.

Evidently, the likely worst-case is encountered on day 1 of injection (when the reservoir is at its lowest depleted pressure). In the experience of the author of this manuscript, initial depleted pressures can be as low as 5 bar and, in conjunction, with poor injectivity reservoir and high pressure injection, this can lead to temperature drop across the formation of more than 20°C. Also, this is the case when injection temperatures is cold and the geothermal gradient is not sufficient to warm up the fluid down the well to overcome the JT cooling across the formation. In order to manage these risks, heating (tophole or downhole) can be provided (with the associated penalties on CAPEX, OPEX and carbon emissions). If hydrate is formed downhole, the injectivity is reduced leading to reduced flowrates. Under this low flow scenario, a higher fluid temperature in the well and downhole is expected. This higher temperature together with heating from the reservoir can melt hydrates that were previously formed. Another hydrate management strategy is to batch inject thermodynamic hydrate inhibitor (typically MEG) to flush and wash the well and any residual brine.

Alternatively, consideration should be given to enhance permeability of the reservoir, hence reducing the pressure drop across the formation and, as consequence, JT cooling. This author has observed good permeability reservoirs with less than 5 bar pressure leading to very mild (or at least manageable) JT cooling (no more than 5°C cooling).

In summary, understanding and mitigating Joule-Thomson effects (e.g., low temperature management) is key when designing CO₂ injection into depleted reservoirs. It follows that predicting accurately the thermodynamic behavior of the fluid is essential.

The opportunity of reusing offshore infrastructure that otherwise would need to be decommissioned after end of life-of-field

CCS transport networks will require the development and construction of infrastructure, including onshore and offshore pipelines, compressor stations, platforms, injection wells and subsea manifolds. This will require high upfront investments. As discussed earlier, depleted gas reservoirs have the main advantage that, in general, they have an infrastructure in place (pipelines, platforms and wells) that could be repurposed for CO₂ service. The rationale to reuse these installations is two-fold. First, decommissioning costs of ageing assets can be delayed. Second, the considerable cost and carbon footprint reductions of CCS projects (at least at early stages of the project enabling large-scale deployment and contribute to net zero targets). However, it is crucial that the integrity of the installations with the change of service is fully demonstrated. It is noted that repurposing existing offshore infrastructure is not only constrained to CCS. It has also been considered to repurpose it for offshore wind farms or geothermal energy production.

Depending on the local regulations, one of the key issues to bear in mind is the period between the end of economic life and the start of CCS operations. Depending on the duration of such "waiting" period^[2] (which will include consenting applications and engineering design activities), it may be necessary to apply alternative temporal regulations and preservation strategies to maintain the integrity of the infrastructure (Drankier and Gazendam, 2021). Nevertheless, it has been noted that decommissioning regulations may need to be reviewed to enable reuse of existing infrastructure (Banet, 2020).

As mentioned earlier, upcoming industrial CCS clusters are taking advantage of existing offshore infrastructure, and this is being repurposed in different fashions. For example, HyNet will repurpose most of the existing offshore infrastructure (at least at early stage of CCS operations), including pipelines, platforms, and wells. While the offshore pipelines are deemed suitable for CO₂ service (subject to final inspection), sidetracking of the existing wells planned and redevelopment of the platforms is expected (Becker et al., 2021). On the other hand, the V Net Zero project is considering (at Concept stage) to repurpose an existing offshore pipeline and install a new platform and injection wells.

Wells

In theory, any type of well (exploration, appraisal, production, gas-lift, etc.) could be repurposed to inject CO₂ (or for monitoring) but their true potential is dependent on their current integrity and fitness for the new fluid service. This is the main risk with reusing existing wells. Plugged and abandoned wells may not be suitable for injections due to the possible lack of integrity data. Preserving the integrity of the well (repurposed or new) is crucial for a successful CCS development. A review of integrity issues in operating CO₂ injecting wells can be found in [Iver et al. \(2022\)](#).

In 2020, the results of a consultation, carried out by the UK Department for Business, Energy and Industrial Strategy (BEIS, 2020), revealed that the reuse of wells will be less common as they would need to be situated in the correct location (for an efficient and safe flow of the CO₂ plume or to avoid fractures and leakage from the store) and meet the required safety standards. Reusing wells will likely require significant workover and recompletion with increasing costs for the CCS project (but lower cost impact than having to drill and complete new wells).

The technical workflow to determine the feasibility of reusing existing wells for CO₂ service consists of several ‘tests’ to which the original well design shall be compliant. Briefly, the minimum technical criteria that the well design is expected (but not limited) to pass are:

- **Material compatibility:** whether the materials of construction are suitable for CO₂ environment as well as other potential corrosive or aggressive impurities to the metallurgy (strong acids, sulfur and nitrous compounds, H₂, etc.). Also, assessment of non-metallic components such as elastomers (polymer sealings) should be conducted.
- **Structural integrity:** verify the mechanical response of the well components (wellhead, casings, tubing, annulus, cement, etc.) to operational conditions. This will include response to cold start-up, induced Joule-Thomson due to well choking, thermal contractions along the well, thermal and pressure cycling, venting, well kill for abandonment, cement degradation, annulus fluid integrity, etc.
- **Well barrier elements integrity:** verify that adequate primary and secondary safety barriers are in place and work correctly. This will include assessment of downhole safety valves, chokes, actuated valves, annulus valves, shoes, packers, etc. Migration/leakage of CO₂ from the reservoir to the annulus is also part of the assessments.
- **Flow Assurance:** verify the thermal-hydraulic behavior against the required injection pressures and design pressure and temperature. Also, this will help to determine the hydraulic capacity and verify if additional wells are required if a target flowrate has to be met.

It is understood, at the time of writing, that there are no standard procedures to qualify existing oil and gas wells for CO₂. The recent REX-CO₂ project has developed a screening tool/workflow to determine the feasibility of reusing existing wells ([Pawar et al., 2021](#)). The tool is based on applicable standards, regulatory requirements and information from projects that have considered repurposing gas wells. In [Raza et al. \(2017\)](#), a guideline for assessing the suitability of well in depleted reservoirs for CO₂ is presented. The guideline aims to support the selection of wells with low risk of leakage and high chance of injectivity. The workflow considers wellbore leakage, near-wellbore characteristics (porosity, permeability, thickness, rock strength) as well as the caprock fracture gradient.

Guidelines and standards for CO₂ well integrity and design. Table 1 suggests technical publications (guidelines, standards, reports, best-practices), in addition to the publications referenced in Section 2.1, that may assist with the design of CO₂ injection wells. It is noted that this list might not be exhaustive/complete and other documentation may be available elsewhere.

Table 1—Documents related to design of CO₂ wells

ISO 27914:2017 Carbon dioxide capture, transportation, and geological storage — Geological storage
US EPA Underground Injection Control (UIC) Class VI – Wells used for Geologic Sequestration of CO ₂
DNVGL-RP-J203 Geological Storage of Carbon Dioxide
DNV CO2WELLS Guideline for the risk management of existing wells at CO ₂ geological storage sites
IEAGHG 2018-06 Re-Use of Oil & Gas Facilities for CO ₂ Transport and Storage
D2.1 REX-CO ₂ Current state-of-the-art assessments and technical approach for assessment of well re-use potential and CO ₂ /brine leakage risk
Porter et al. Converting Wells in a Mature West Texas Field for CO ₂ Injection, SPE 20702, 1990
Haigh. Well Design Differentiators for CO ₂ Sequestration in Depleted Reservoirs, SPE 124274, 2009
Duguid et al. Baseline integrity property measurement of legacy oil and gas wells for carbon storage project, Greenhouse Gas Sci. Technol., 2017
Duguid et al. CO ₂ Well Construction: Lessons Learned from United States Department of Energy Sponsored Projects, GHGT-14, 2020

Platforms

According to a recent consultation (BEIS, 2020), the condition of platforms deteriorates quickly after production operations cease. Very few respondents to the consultation recommended their reuse and, if it is possible to repurpose platforms, they will require significant modifications. It was also suggested that distinction should be made between the platform and jacket structures because it may be possible to remove the topside and retain the jacket structure, retrofitting a smaller, fit-for-purpose module.

The remaining life expectancy of platforms will also be a key risk in assessing their feasibility for repurposing. Many platforms were designed for the initial field development, while many others had modifications to expand their original design (thereby adding loads). If equipment can be removed from an existing platform (thereby reducing the loading on the structure), it may be possible to extend further the service life of a platform that has already exceeded its original fatigue design life.

Thus, if platforms are to be reused, they will need to be upgraded to suit the new service. This may include (but not limited to) replacing piping, installing filters and advanced control, monitoring and metering systems. Space and weight restrictions will also play crucial factors in the feasibility of reusing platforms. This is important if, for example, wellhead compression or any other major equipment is required. Other factors to consider when assessing platform reusability are power requirements, available well slots, and new built infrastructure tied back to an existing platform (Sachde et al., 2019). It follows that if platforms are not suitable for reuse and need to be decommissioned this will impact the ability to reuse existing wells.

Pipelines

Reusing existing oil and gas pipelines to transport CO₂ can provide significant cost reductions to CCS projects. In principle, there are no serious technical implications in reusing existing offshore pipelines for CO₂ service. It is envisaged that existing pipelines have the greatest potential to be repurposed for CCS transport network (BEIS, 2020). However, there are key technical considerations/risks that should be identified if a pipeline is a candidate to be repurposed.

At the time of writing this manuscript, there is not a standard procedure or guideline to evaluate the potential of converting existing pipelines to CO₂ service. The Energy Institute will be publishing a guideline for this aim in 2022. In the meantime, the following technical criteria (as a minimum) is expected to be part of any workflow to assess/screen the potential of repurposing pipelines for CO₂:

- Flow Assurance: estimate the hydraulic capacity of the pipeline and verify that required operating pressures do not exceed original design pressures. This is discussed further in Section 2.3.1.
- (Internal) Corrosion risk: identify the potential of carbonic corrosion if there is presence of water free phase. Corrosion rates can be high (several mm/y). Moreover, consideration to other damage

mechanisms such as H₂S corrosion, stress cracking, O₂ corrosion, etc., should also be part of the risk assessment. More discussion can be found in Section 2.3.2.

- Mechanical design: Of particular interest in CO₂ pipelines is a phenomenon known as running ductile fracture. Here, a small initial fracture can start running along the pipeline in both directions; in the worst case, this may potentially split hundreds of meters of pipeline, causing large and, potentially, dangerous leaks. Evaluations must be carried out to ascertain if the design toughness requirement of a reused pipeline would be suitable for transporting, in particular, dense phase CO₂.
- Risk register and assessment: A robust risk identification, scoring tool and mitigation measures shall be developed from initial stages in the project. The register should contain not only risks associated to the particular asset but also lessons learned from analogous projects. Risk assessment and management are essential parts of CCS developments in order to provide answers for the uncertainties and assure the safe operation of the CCS network. It is noted that, whenever possible, risk identification should reach beyond engineering/technical risks also addressing commercial issues to social perception matters.

Flow Assurance. From the Flow Assurance perspective, the following drivers are key when consider the viability of existing infrastructure for CCS operations:

- Original design/maximum allowable pressures
- Fluid specification and expected impurities profile in the stream
- Phase in which CO₂ will be transported in the pipeline
- Minimum design temperature

When "CO₂" does not mean 100% pure CO₂

Capture CO₂ from flue or natural gases is unlikely (never) pure. Certainly, high purity CO₂ could be achieved but incurring in high energy penalties. In any case, the concentration and type of impurities expected from various CO₂ sources connected to the industrial CCS hub will differ depending on the technology that is producing CO₂ and the capture technology.

There is a need to establish a common specification for the impurities within the CO₂ stream and to design cost effectively a pipeline system capable of handling as wide a range of impurities as possible without compromising the integrity of the existing system. Several CO₂ specifications have been developed over the years (Murugan et al., 2020) but, at the time of writing this manuscript, there is not a "standard" specification for industrial CCS hubs. Although these specifications serve as guidance for the design of the components of the CCS system, each development will develop its own custom-specification that is appropriate for its own characteristics. Thus, each emitter will be discharging its CO₂ at different compositions but within the required specification. Therefore, the mixture to be transported could consist of all the possible compounds associated to each capture and purification technology (i.e., solvents or other chemicals). It is realistic to expect that more than 20 impurities may be present in a CCS mixture (Luna-Ortiz, 2021). Understanding the thermo-physical and transport properties of the CCS mixture is fundamental (Nazeri et al., 2021). Even small amounts of impurities could significantly change the thermodynamic behavior with respect to the one of pure CO₂. For example, very small amounts of glycols could induce a, potentially, corrosive aqueous phase at low temperatures. Failure to anticipate this could lead to catastrophic consequences.

Supercritical CO₂ is known to be an excellent solvent and medium for several chemical reactions (for the so-called 'green chemistry') (Licence et al., 2003; Han and Poliakoff, 2012). In the context of CCS, chemical reactions of various impurities (mainly NO_x, SO_x) in supercritical CO₂ creating corrosive products (aqueous phase) have been studied (Morland et al., 2021; Morland et al., 2019). Consequently, there is a possibility that further purification of the CO₂ may be required (thus increasing costs and energy penalties)

to avoid any integrity risk. Moreover, it is possible that many other chemical reactions, amongst the different impurities, may occur in the pipeline (essentially a very long tubular reactor). This may create additional species (or increase the content of existing impurities) leading to further risks that should be anticipated. Typical reactions that it could be expected to occur will be from reactants used in the capture plants (e.g., production of carbamates from reactions of CO₂ with ammonia or amines (Versteeg and van Swaaj, 1998; Rochelle, 2009). At this stage, this is purely conjecture. Reactivity and spontaneity (chemical pathway feasibility) of any potential chemical mechanism and its kinetics should be investigated.

Mapping the operating limits of existing pipelines for CO₂ transport

Existing pipelines have an original design pressure which should be honored (if the pipeline has been derated during production life, there is a possibility to uprate back the pressure). Thus, the hydraulic capacity of the pipeline (maximum flowrate) will be constrained with operating pressures always below the design pressure.

For optimal transport of CO₂ in dense/liquid phase, the minimum operating pressure should exceed the critical pressure for CO₂ mixture (hence, the need to clearly identify the impurities in the mixture as the critical pressure depends on the composition). This is to avoid phase splitting (liquid-gas) in the pipeline. It is generally agreed that two-phase flow should be avoided due to potential difficulties in operation of process equipment. While two-phase flow in hydrocarbon systems is well established, this is still not the case in CO₂ systems (at least not in large-scale systems). Until the behavior of multiphase CO₂ flow is better understood and predicted with simulation tools, a key design premise is to operate in single-phase. Currently, there is ongoing research activity (Yan et al., 2021; Andersen et al., 2021) but there is not sufficient evidence to advance the design of CCS transport systems in multiphase flow mode.

If the existing pipeline is not sufficiently rated for transportation of CO₂ in dense/liquid phase, then transportation in gas phase will remain as option. Evidently, for the same pipe diameter, higher flowrates can be achieved in dense phase than in gas phase. Consequently, transportation in gas phase will be significantly constrained and upgrades (pipeline replacement) may be required to accommodate higher flowrates.

Recent studies (Cauchois et al., 2021; Carbon Limits and DNV, 2021) have identified the potential to repurpose existing pipelines in Europe for CO₂ transport. The analyses focused on the hydraulic capacity and transport costs in comparison to new built pipelines. The presented case studies show significant cost savings when reusing existing infrastructure. These screening studies/methodologies are useful for identifying candidate assets for reuse. Once a candidate infrastructure is selected additional feasibility studies should be carried out to take into consideration composition variability, terrain/elevation changes, thermal dependences, etc. In Seevam et al. (2010), a case study is presented in which it is shown, for example, how different compositions will impact the feasibility of an existing pipeline to transport CO₂ in different phases.

System constraints/operating maps can be developed for a candidate infrastructure. The operating envelope provides a tool to visualize the complex interactions between different components of the transportation network. In addition, it enables identification of required modifications to debottleneck the network for future expandability. The operating map should be produced with rigorous thermo-hydraulic modelling and thermodynamics taking into consideration phenomena such as Joule-Thomson cooling due to gas expansion along the flowlines.

Figure 2 shows the operating envelope for an existing infrastructure showing the different pinch points and required modifications as pressure and/or flowrates increase. In this system, operation in gas phase is carried out throughout the project life (onshore and offshore systems). First, upgrading the capacity of the offshore system (new pipeline size and new ratings) to extend operation in gas mode (no intermediate compression) once the maximum operating pressure is reached in the existing system. The offshore wellhead compression (this has the advantage of significant energy savings as it eliminates the need for significant post-compression cooling and offshore heating and assumes there is space in the existing platforms) will be installed once the upgraded system reaches its maximum capacity. In this system, ‘minor’ retrofit of the

existing facilities is required. Finally, the system can be further expanded with additional upgrade of the existing onshore pipeline. With this, the transport network will be ‘ready-for-liquid’ for future operation.

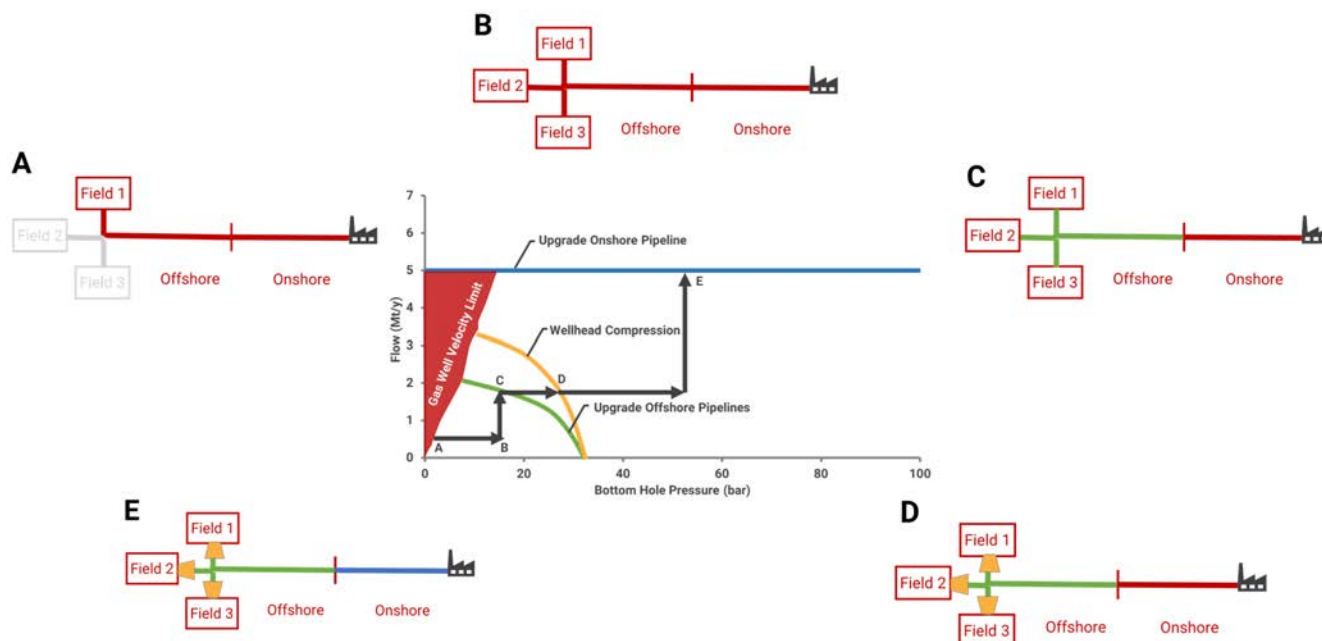


Figure 2—Mapping operational envelope and network evolution of existing infrastructure. A Start of operation with existing facilities. Low pressure (gas phase across system) and low flowrate (single emitter). System is constrained due to high gas velocities inducing excessive Joule-Thomson cooling transgressing original minimum design temperature of the existing system. B With increasing reservoir pressures, lower velocities induce less Joule-Thomson cooling. Gas phase operation across the system. C Flowrate ramp-up (more emitters) triggers upgrade of offshore pipelines (increase size). Gas phase operation across system. D Pressure boosting is required to keep injecting against higher reservoir pressures. Gas phase operation in system. E Upgrade of onshore section to increase overall system capacity. Gas phase operation but system ready for future high pressure liquid operation.

Corrosion and Materials. Unless existing pipelines are made of beskar[3], vibranium[4], or any other real or fictional corrosion-resistant alloy, these will experience accelerated corrosion (as most likely they are stainless steel pipelines) in the presence of free water when transporting CO₂. Hence, the key is to keep the fluid dry under, essentially any operating mode.

The water solubility is strongly affected by impurities in the CO₂ stream. The change in solubility will not be very large if the impurities are inert (not undergoing chemical reactions) and non-condensable (e.g., Ar, N₂, H₂ and CH₄). A much larger solubility change is observed when impurities are undergoing chemical reactions forming new products and phases (as discussed earlier). Aqueous phases can also form more easily when salt, glycols, and methanol (or other strongly associating compounds) are present. These impurities will absorb water and generate an aqueous corrosive phase at water concentrations far below the solubility limits reported for pure CO₂.

Furthermore, consideration to other damage mechanisms (as per API 571) shall be given. These are (but not limited to):

- Carbonic acid corrosion
- H₂S corrosion
- H₂S Sulfide stress cracking
- H₂S Stress-oriented stress cracking
- H₂S Hydrogen induce cracking

- Hydrogen embrittlement
- Top of line corrosion
- O₂ corrosion
- Microbially induced corrosion
- Chloride stress corrosion cracking
- Corrosion fatigue
- Amine corrosion
- Sour water corrosion

Guidelines and standards for CO₂ pipelines integrity and design. Table 2 suggests technical publications (guidelines, standards, reports, best-practices), in addition to the publications referenced in Section 2.3, that may assist with the design of CO₂ transport pipelines. It is noted that this list might not be exhaustive/complete and other documentation may be available elsewhere.

Table 2—Documents related to design of CO₂ pipelines

ISO 27913:2016 Carbon dioxide capture, transportation, and geological storage — Pipeline transportation systems
ISO 27921:2020 Carbon dioxide capture, transportation, and geological storage — Cross Cutting Issues – CO ₂ stream composition
ISO 27918:2018 Lifecycle risk management for integrated CCS projects
DNVGL-RP-F104 Design and operation of carbon dioxide pipelines
DNV CO2RISKMAN Guidance on CCS CO ₂ Safety and Environment Major Accident Hazard Risk Management
World Resources Institute (WRI) CCS Guidelines for Carbon Dioxide Capture, Transport, and Storage
Energy Institute Hazard Analysis for Onshore and Offshore Carbon Capture Facilities and Pipelines – <i>in preparation</i>
Energy Institute Good Plant Design and Operations for CCS installations and pipelines – <i>in preparation</i>
Energy Institute Repurposing and Design Guidelines for Carbon Dioxide Pipelines – <i>in preparation</i>
IEAGHG 2018-06 Re-Use of Oil & Gas Facilities for CO ₂ Transport and Storage
IEAGHG 2013-18 CO ₂ Pipeline Infrastructure
Mohitpour et al. Pipeline Transportation of Carbon Dioxide Containing Impurities, ASME Press

Conclusions

Meeting climate targets without large-scale deployment of CCS will be extremely challenging. While CCS is already being used and will be continued to be used, in oil and gas developments (for the disposal of CO₂ from natural gas processing) worldwide, this hardly has any significant impact into climate targets as emissions from the burning of fuel is not abated.

To accelerate implementation and increase emissions capture rates, industrial clusters with shared transport facilities has been proposed as a cost-effective pathway to enable rapid large-scale CCS deployment (Martin-Roberts et al., 2021). Depleted (or depleting) gas fields with existing infrastructure (pipelines, platforms, and wells) could offer additional savings. For operators, it offers the incentive to delay decommissioning costs. However, it is crucial that the integrity of the installations with the change of service is fully demonstrated and guaranteed. The aim of this paper is to review and discuss the main technical challenges that repurposing pipelines for CO₂ transport pose.

The key technical challenges/risks include understanding that, firstly, CO₂ will not be pure, and that any impurity will change the thermo-physical behavior of the fluid. Secondly, while CO₂ is a relatively benign

fluid for the metallurgy, the various impurities can produce corrosive/aggressive products that could threaten the integrity of the system. Existing systems will have a design pressure rating which will constrain the hydraulic capacity and the phase at which the fluid could be safely and robustly be transported. Failure to early identify (at screening stages of the projects) these risks will be detrimental to the development of the CCS scheme.

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